

# Overview Of Common Cardiac Tests

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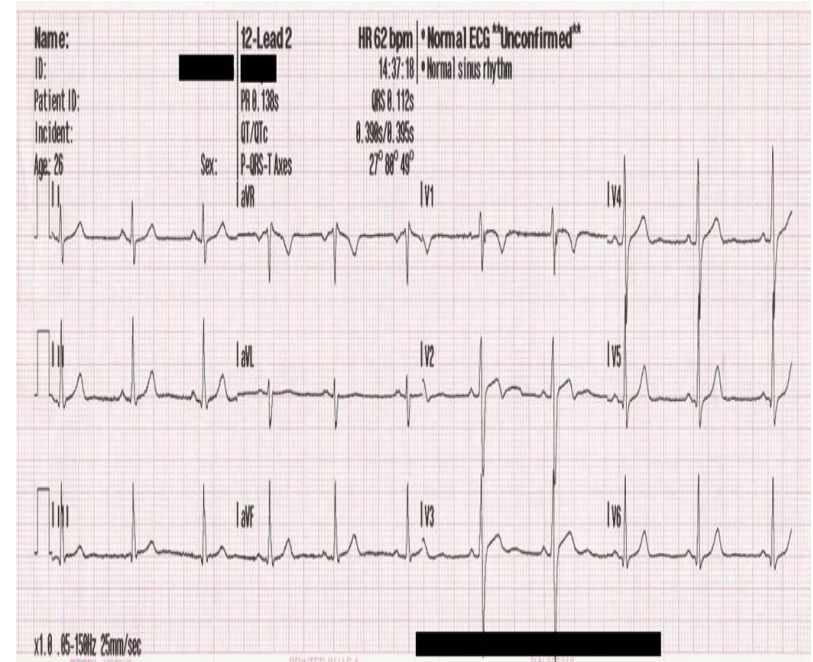
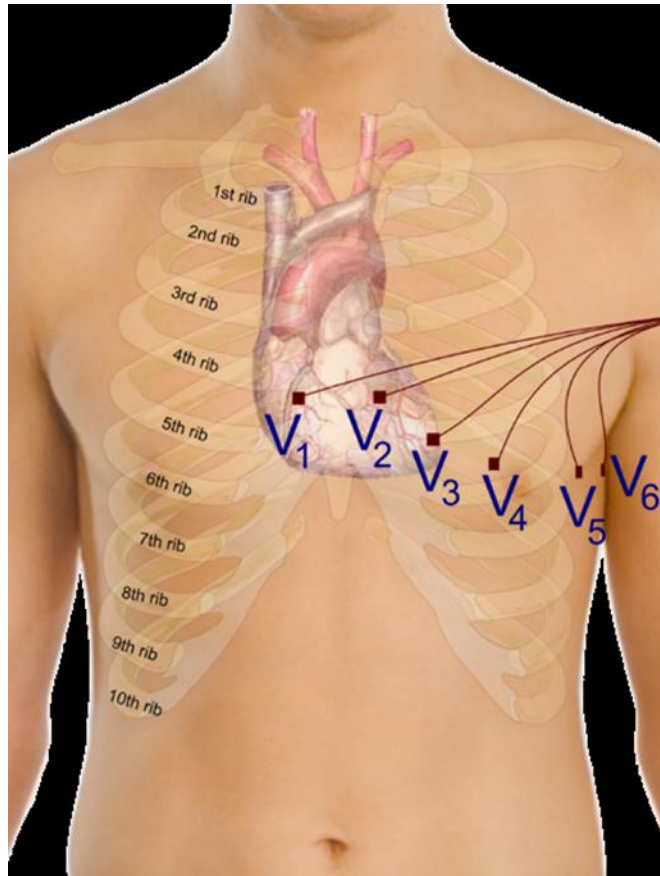
Joanne Gilham RN  
Insurance Nurse Specialist, Unum  
Health Claims Forum  
26<sup>th</sup> February 2015

# Aims And Objectives

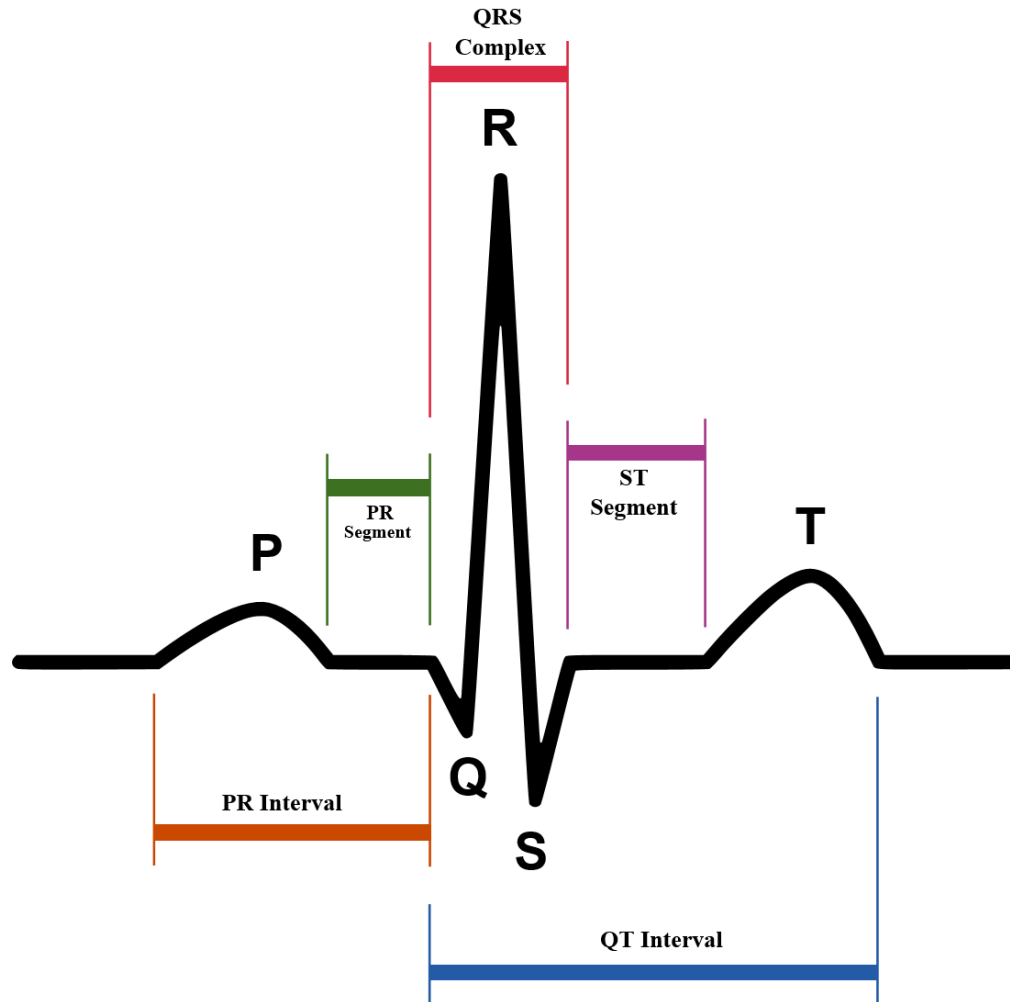
- Overview of commonest cardiac investigations
- Values to look for in test results
- How it equates to day to day functioning

# 12 Lead Electrocardiogram (ECG)

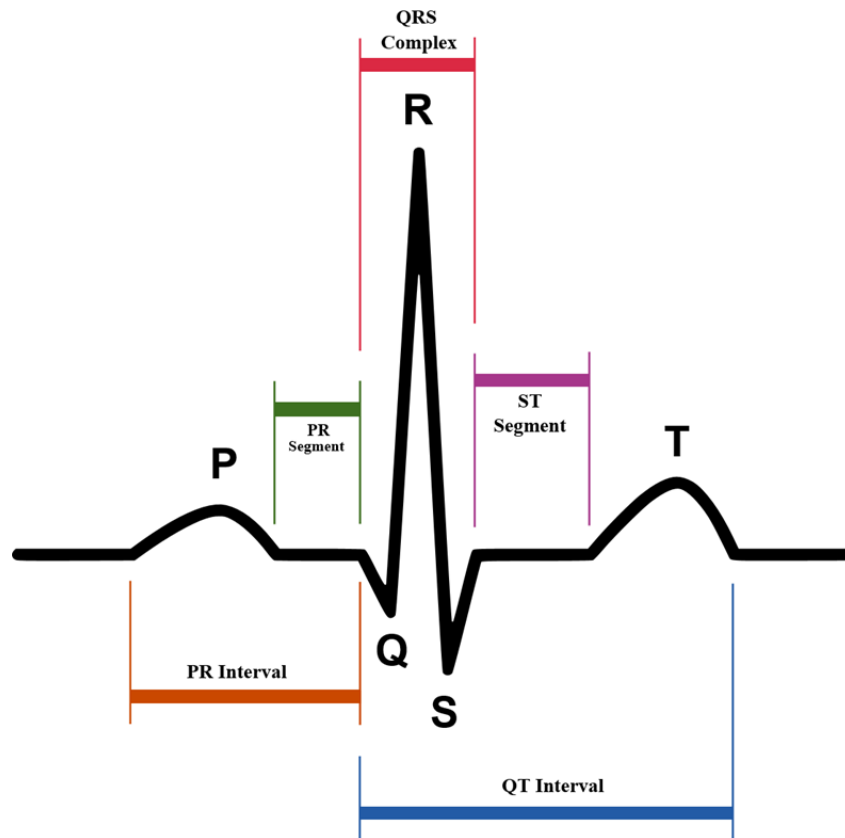
- After baseline observations, should be the first test performed when patient presents with chest pain
- Non-invasive
- Takes around a minute
- Looks at the electrical impulses through the heart
- Can detect rhythm disturbances, acute / previous damage to the heart muscle and conduction difficulties within the heart



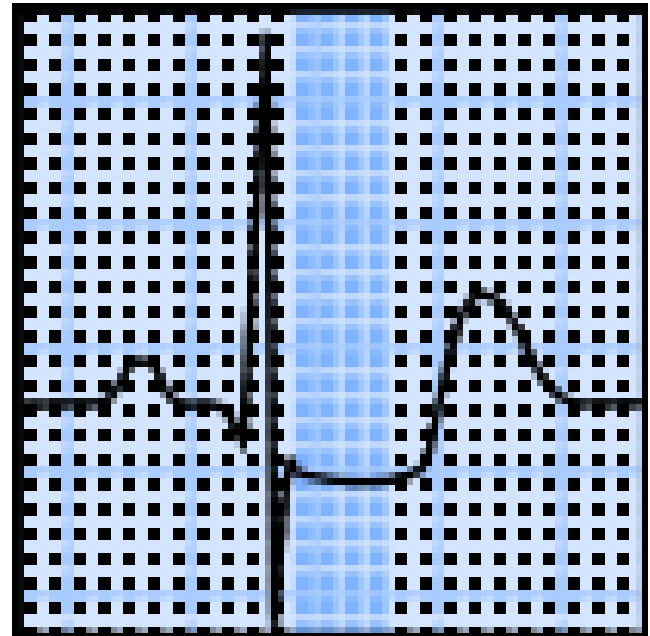
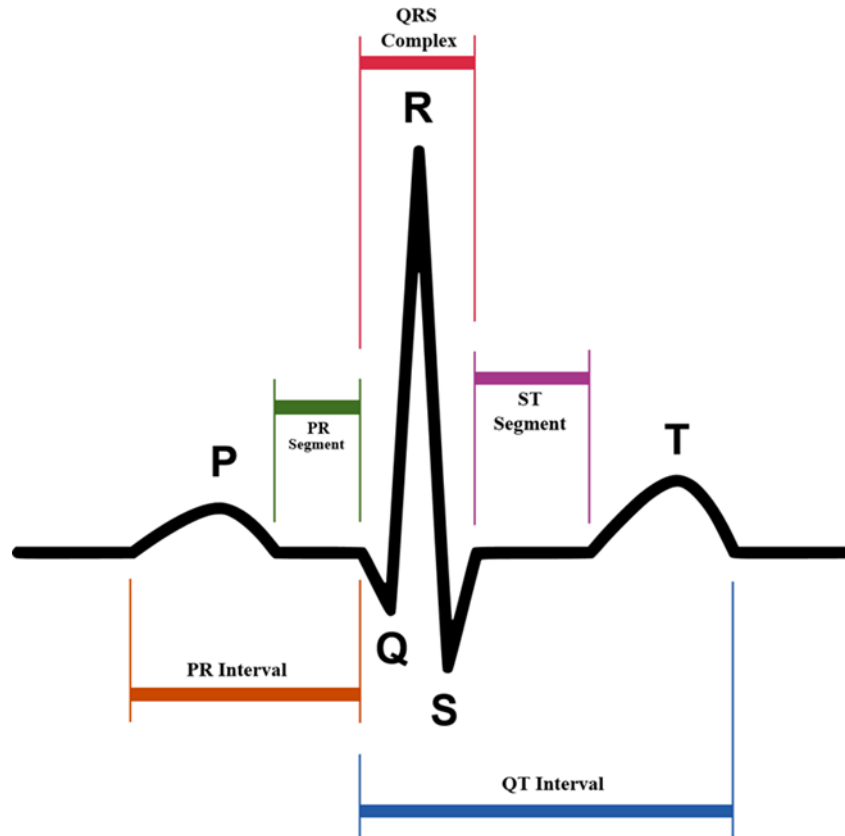
# Normal Complex



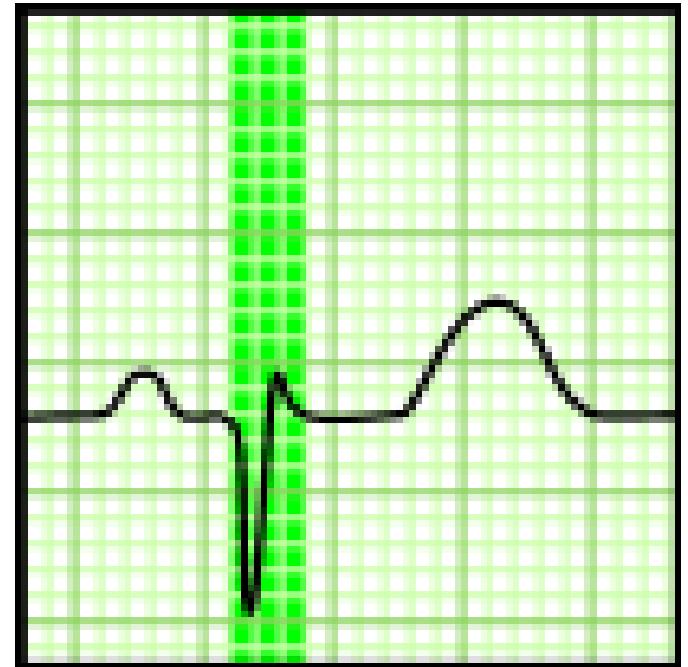
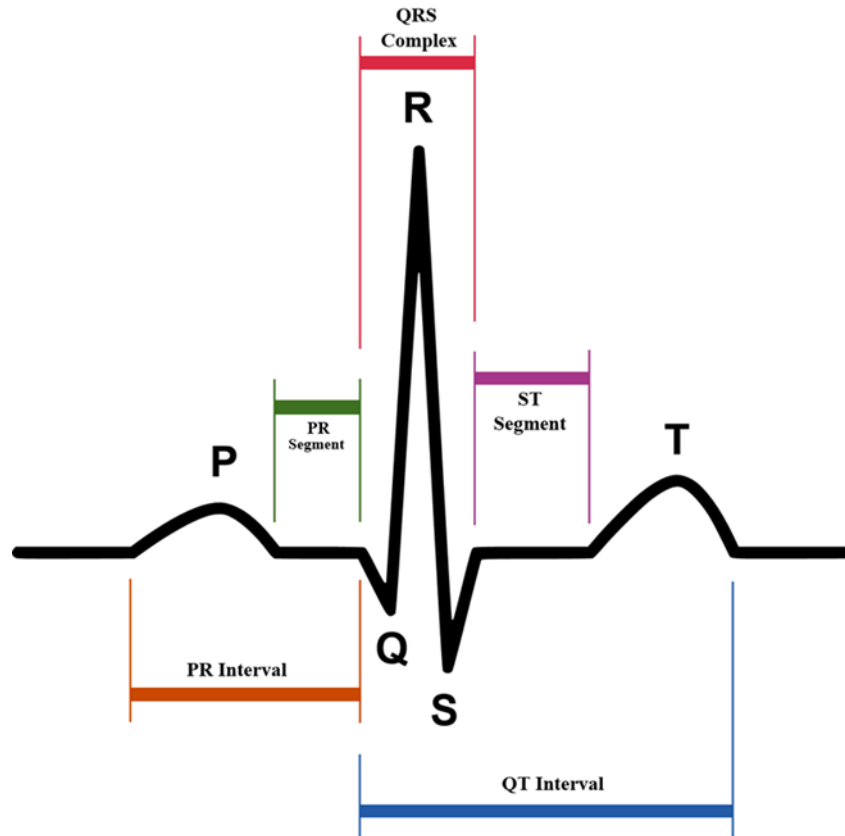
# ST Segment Elevation



# Other Significant ST Segment Changes



# Other Significant ECG Changes





# Exercise Tolerance Test (ETT, Treadmill Test)

- Two week rule to be seen in a rapid access chest pain clinic
- Patient is attached to ECG machine and BP monitor
- Aim is to provoke symptoms and see what is happening at a cardiac level when symptoms present
- Closely medically monitored environment
- Bruce Protocol usually followed
- Slow walking pace on a treadmill
- Three minute increments, pace speeds up, tilt added to treadmill



## ETT cont.

- Need to look at reasons why the test has stopped
- Chest pain
- ECG changes
- Hypotension / Hypertension
  - » Indications that cardiac cause to symptoms
- Shortness of breath
- Cramps
- Fatigue
  - » May not be cardiac reason for stopping test
- Following test, patient remains attached to monitors during a recovery period

# Metabolic Equivalent Task (MET)

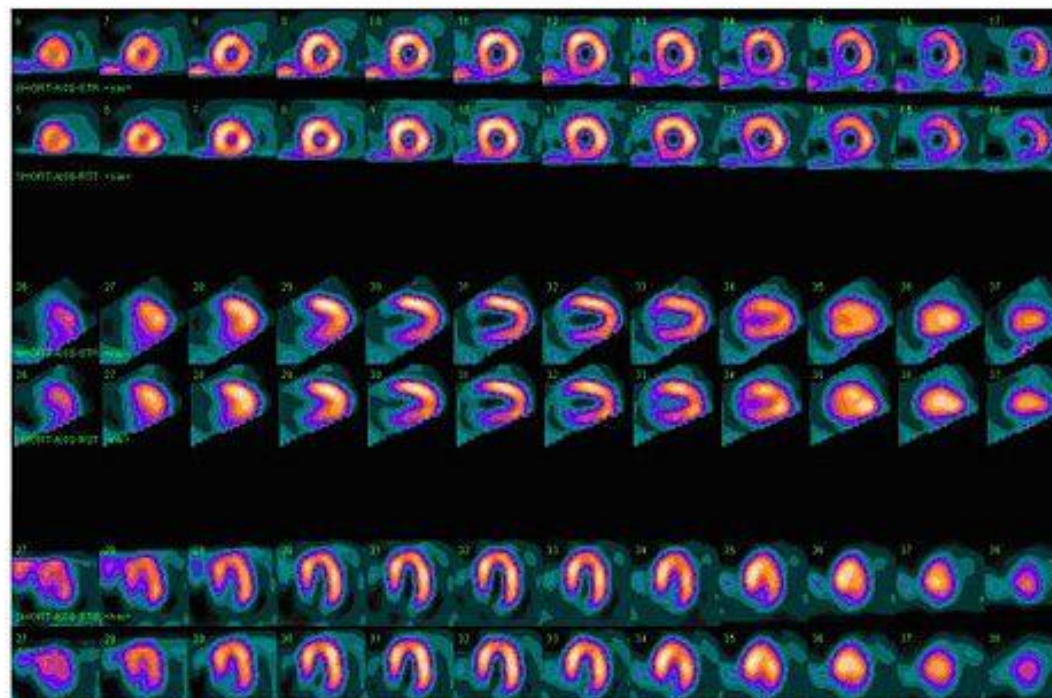
- An average person at rest will use an oxygen consumption of 3.5ml oxygen per kg X minute. This equates to 1 MET
- The first stage of the Bruce Protocol equates to around 5 METs
- Every stage the increase in physical activity equates to an increase in 2 METs

# METs cont.

- **Example of Activities**
- **Light (<3.0 METs)**
  - Slow walking
  - Sitting using a computer
  - Ironing
- **Moderate (3.0 – 6.0 METs)**
  - Brisk walking
  - Mowing the lawn
  - Heavy Cleaning (e.g window washing)
- **Vigorous (>6.0 METs)**
  - Sport such as playing football or singles tennis
  - Cycling quickly (14 – 16 mph)
  - Hiking

# Myocardial Perfusion Scan (Thallium Scan)

- Small amounts of radioactive dye (radionuclide) used to assess blood flow to the myocardium
- Usually performed twice, once at rest and once after a period of light exercise
- Radionuclide travels to the heart muscle where normally perfused heart muscle will absorb the substance, whilst areas with poor blood supply will not absorb it as well
- More gamma rays are emitted from well perfused areas than the poorly perfused areas
- A gamma camera is used to pick up the signals and scans for around 15 – 30 minutes
- Assess heart muscle only, not coronary arteries
- Report will talk about viable segments
- 89% sensitivity for detecting cardiovascular disease



# Cardiac Angiogram

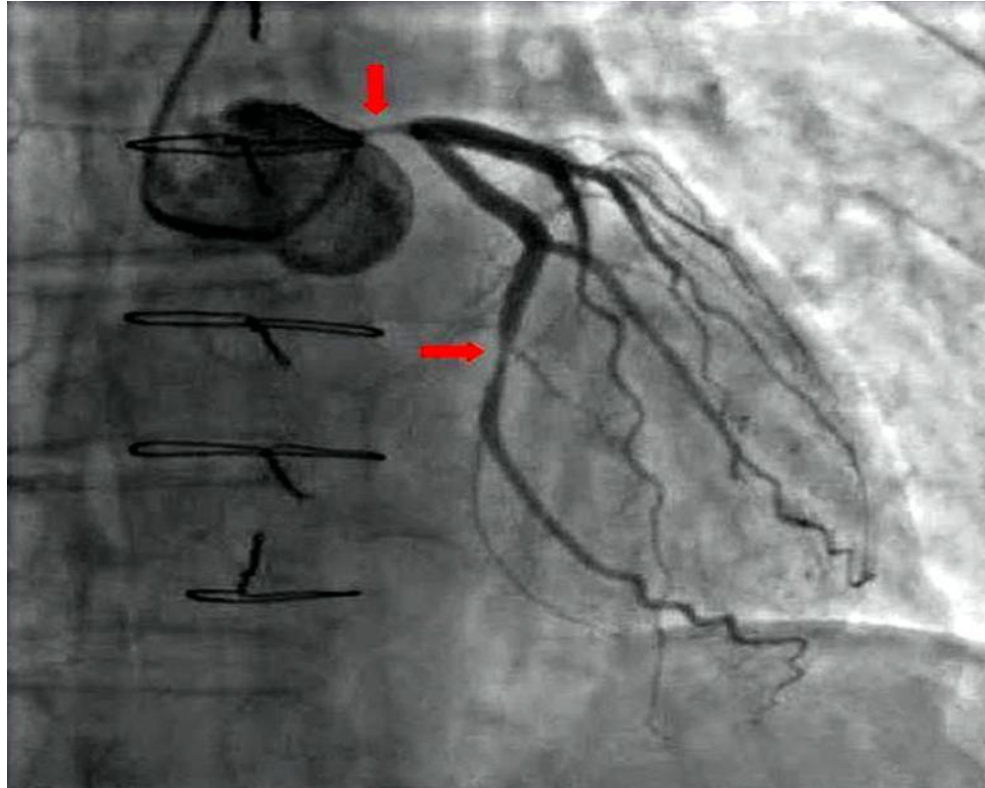
- Invasive cardiac investigation that looks at the blood flow through the coronary arteries
- In last 15 years, has become more widely available in district general hospitals
- Considered one of the best tests for diagnosing coronary heart disease
- Patient is awake for procedure that lasts around 30 – 45 minutes
- Can be done on an outpatient basis
- Involves insertion of a thin catheter into an artery, usually in groin or arm, which is fed through to the heart
- Radioactive dye is injected through this and an x-ray machine sitting above the patient's chest will record pictures of the dye as it is pumped through the coronary arteries
- The aim is look for any blockages within the coronary arteries



# Angiogram Laboratory



# Cardiac Angiogram

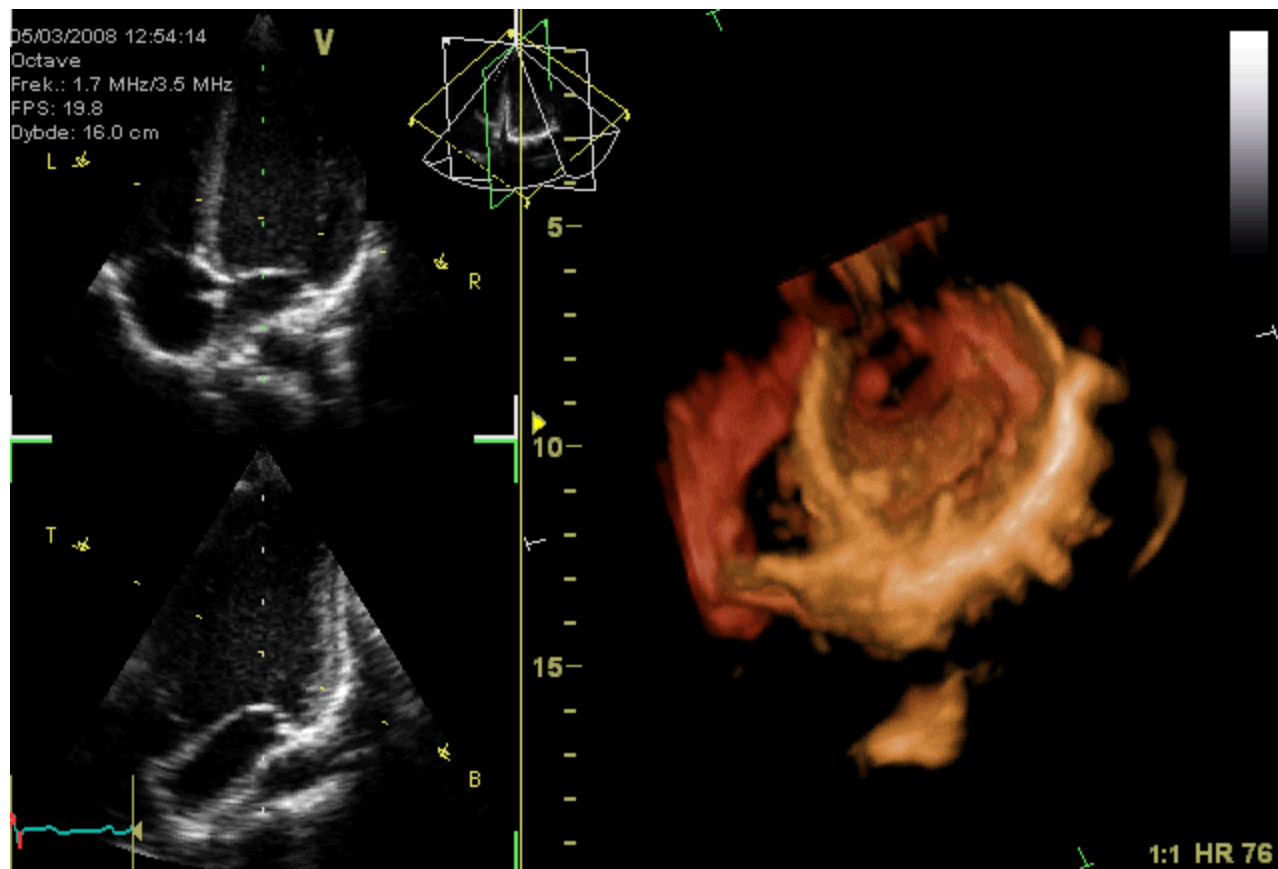


- Maria A Pantaleo, Anna Mandrioli, Maristella Saponara, Margherita Nannini, Giovanna Erente, Cristian Lolli and Guido Biasco : *Development of coronary artery stenosis in a patient with metastatic renal cell carcinoma treated with sorafenib*. BMC Cancer, 2012, 12:231 [doi:10.1186/1471-2407-12-231](https://doi.org/10.1186/1471-2407-12-231) Published: 11 June 2012

# Echocardiogram

- Non invasive ultrasound scan that looks at the structure of the heart, functioning of the valve and the pumping action of the heart
- Commonly performed post myocardial infarction and in patients with suspected or known heart failure
- Lasts around 15 – 30 minutes

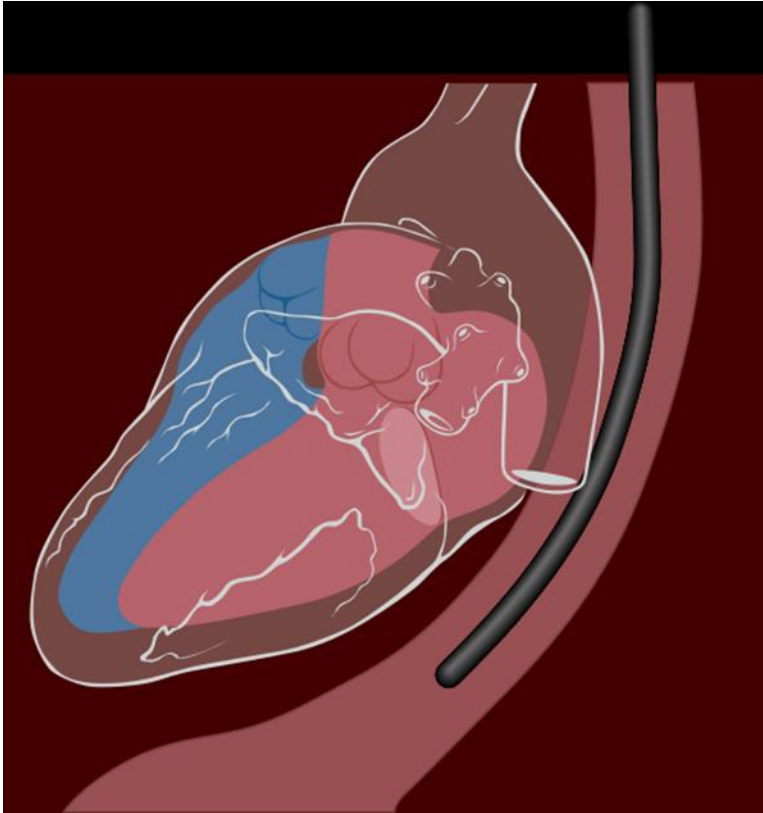




# What We Want To Know In An Echocardiogram Report

- What are the valves doing?
  - Is there mild / moderate / severe regurgitation?
  - Signs of valve prolapse?
  - Are there any vegetation's growing on the valves?
- How is the wall of heart looking?
  - Is a part of the wall of the heart classed as hypokenetic?
  - Any mention of hypertrophy?
- What is the ejection fraction?
- What percentage of the blood in the left ventricle is being pumped around the body in one heart beat?
- Is this any different to previous echocardiograms?

# Transoesophageal Echocardiogram (TOE)



- A small probe is passed down the oesophagus so the heart can be visualised clearer
- Often light sedation is required
- Can give a clearer picture of the valves
- Often used in obese patients
- The report will contain the same information as a standard echocardiogram report

- Patrick J. Lynch, medical illustrator; C. Carl Jaffe, MD

# Cardiac MRI Scans

- Scan that allows assessment of cardiac anatomy, the functioning of the heart including valves, looks for signs of ischemia and infarction, presence of tumors and degree of heart failure
- Non-invasive
- Does not use radiation
- Takes around thirty to ninety minutes
  
- However, not all hospitals have an MRI scanner or practitioners trained to undertake cardiac MRI scans
- Predicted that in the near future that this will become the gold standard cardiac investigation



Jan Ainali





# Any Questions?

